

Kanu Madhok

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PROFESSIONAL EXPERIENCE

Walmart Data Ventures

Chicago, IL · Sep 2024 – Present

Senior Data Analyst

- Designed and shipped a production self-service analytics agent over 67 BigQuery tables — a multi-agent architecture with planner / executor / reviewer / critic sub-agents and context-isolated validation (Context Researcher → SQL Drafter → Validator → Devil's Advocate), governed by 8 deterministic guardrail rules and a 23-case golden-query evaluation suite. Approved by Sr. Director and product leadership and currently the only AI skill in active use by business teams on Data Ventures. Eliminated the analyst as the bottleneck for routine asks.
- Built a hybrid orchestrator on top of the agent above. Routes business questions through two backends in parallel: a CubeJS semantic-layer skill (co-built with the DS and product team) and the context-engineered agent itself. Compares outputs, reasons about disagreements, and feeds discrepancies back to refine the semantic layer's definitions over time.
- Built an autonomous Jira ticket-resolution agent: 6-gate triage, MCP-driven context gathering, FAISS+BM25 retrieval over 11,000 historical SQL queries, self-healing execution with a 10-retry budget. Closed 400+ data requests across 8 categories over 15 months — cut turnaround from 30–60 minutes per request (and up to a full day for complex ones) to under 10 minutes each. Stakeholders never knew an agent was answering them.
- Replaced ~20 distinct recruitment workflows I'd been running by hand 1–2× weekly with end-to-end automation on a 68-table BigQuery platform. Stratified-sampling pipelines recruited 280,000+ panelists across 29 categories with zero duplicates. A recurring 3-hour task now takes ~3 minutes of setup.
- Built a headless KPI monitor on a 6-hour cron that detects threshold breaches, runs root-cause analysis, and posts Slack alerts; state persists to BigQuery so the agent has no LLM-memory reliance. Designed a v2 architecture as a 3-tier event-driven platform (reactive events, threshold KPIs, PM-owned themes) self-configuring via a BigQuery event registry. New event types ship without code deploys.
- Delivered 5 Power BI dashboards ranked top 1.4% of Walmart's 207,000 dashboards (1,064 views, 22 users); cut data-request turnaround from 1–24 hours to 5–10 minutes; Making a Difference Award.

University of Chicago — Data Science Institute

Chicago, IL · Jan – Sep 2024

Graduate Researcher, Data & Democracy Initiative

- Persona-driven multi-LLM (OpenAI + Gemini) chatbot for a 500+ participant donation experiment; controlled A/B tests persuaded 15% to donate to an opposing cause.

FTI Consulting — UChicago MSADS Capstone

Chicago, IL · Jan – Aug 2024

Capstone engagement with FTI Consulting, a global financial & business advisory firm

- Built a RAG-to-proposal generator that automated drafting of client engagement proposals for FTI's advisory teams; deployed on AWS EC2, cut proposal turnaround ~30%; Best in Show — UChicago MSADS Capstone.

Innovare (EdTech Startup)

Chicago, IL · Oct 2022 – Mar 2023

Data Scientist

- Logistic-regression risk model + Looker dashboards for 10+ districts (5,000+ students); retention +7%, failures –15%.

SELECTED PROJECT

- Rapidly prototyped and live-deployed a text-to-SQL copilot (Streamlit + LangChain + Gemini 2.5 + RAG over historical queries, BigQuery execution, auto-visualization): live demo — <https://sql-rag-frontend-simple-481433773942.us-central1.run.app/>

SKILLS

Agentic AI: multi-agent architecture (planner / executor / reviewer / supervisor patterns), sub-agent orchestration, MCP (Model Context Protocol) server design, tool- & function-calling agents, memory-enabled agents, context & prompt engineering, event-driven agent workflows, headless cron agents

Evaluation & reliability: golden-query evaluation suites, deterministic guardrail rules, offline testing of agent flows, output consistency checks, self-healing execution, decision logging & monitoring

RAG & retrieval: FAISS + BM25 + RRF, Pinecone, ChromaDB semantic memory, multi-model routing across providers

LLMs & frameworks: Claude (Opus / Sonnet / Haiku), Gemini (2.5 Pro / Flash), OpenAI, LangChain, Claude Code (daily — via Wibey, Walmart's internal Claude Code fork)

Engineering & delivery: Python, FastAPI, Cloud Run, BigQuery scheduled queries, Slack webhooks, cron, Playwright SSO, JSONL state stores

Data & viz: BigQuery, dbt, SQL, PySpark, Airflow, Databricks, MySQL; Power BI (DAX, XMLA TMSL), Tableau, LookML, Streamlit, Matplotlib, React Recharts

EDUCATION

The University of Chicago — M.S., Applied Data Science (GPA 4.0)

Chicago, IL

Loyola University Chicago — B.S., Information Systems & Economics (GPA 3.5)

Chicago, IL